Bigmart Sales Data

This is short summary about my python + notebook knowledge, where I tried to read csv data in google colab and try some scripts of code to analyze the data. Here, I review few rows of dataset from top and try to learn about columns. I also try to summarize dataset by Item weight, Outlet_Size, count number of items in each Item_Type and Outlet_Size.

1. Reading csv file-

Code:

```
[6]
✓ 0s
import pandas as pd
data = pd.read_csv('/bigmart_data.csv')
```

- **pandas** library is imported as **pd**
- read csv file as **DataFrame** and **data** is named.

2. Review Dataset

Code:

```
[8]
✓ 0s
data.head()
```

- Use **head()** function to review first 5 rows of dataset,

Output:

	Item_Identifier	Item_Weight	Item_Fat_Content	Item_Visibility	Item_Type	Item_MRP	Outlet_Identifier	Outlet_Establishment_Year	Outlet_Size	Outlet_Location_Type	Outlet_Type	Item_Outlet_Sales
0	FDA15	9.30	Low Fat	0.016047	Dairy	249.8092	OUT049	1999	Medium	Tier 1	Supermarket Type1	3735.1380
1	DRC01	5.92	Regular	0.019278	Soft Drinks	48.2692	OUT018	2009	Medium	Tier 3	Supermarket Type2	443.4228
2	FDN15	17.50	Low Fat	0.016760	Meat	141.6180	OUT049	1999	Medium	Tier 1	Supermarket Type1	2097.2700
3	FDX07	19.20	Regular	0.000000	Fruits and Vegetables	182.0950	OUT010	1998	NaN	Tier 3	Grocery Store	732.3800
4	NCD19	8.93	Low Fat	0.000000	Household	53.8614	OUT013	1987	High	Tier 3	Supermarket Type1	994.7052

3. Summary Table

Code:

```
[32]
✓ 0s
pivot = pd.pivot_table(data, index=['Item_Type'], values="Item_Weight", aggfunc= 'mean')
pivot = pivot.rename(columns = {'Item_Weight':'Mean_Weight'})
pivot
```

- **First line of code**, creates a summary table that shows the **average mean** of **Item_Weight** for each category in the **Item_Type** column.
- **Second line of code,** rename the column name from **Item_Weight** to **Mean_Weight**
- **Third line of code,** pivot is showing a summary table of dataset

Output:

Item_Type	Mean_Weight
Baking Goods	12.277108
Breads	11.346936
Breakfast	12.768202
Canned	12.305705
Dairy	13.426069
Frozen Foods	12.867061
Fruits and Vegetables	13.224769
Hard Drinks	11.400328
Health and Hygiene	13.142314
Household	13.384736
Meat	12.817344
Others	13.853285
Seafood	12.552843
Snack Foods	12.987880
Soft Drinks	11.847460
Starchy Foods	13.690731

Code:

[37]
✓ 0s

```
pivot_outlet_size = pd.pivot_table(data, index=['Outlet_Size','Outlet_Type'], values='Item_MRP', aggfunc='mean')
pivot_outlet_size
```

- A table summary of **average Item Price** by **Outlet_Size** and **Outlet_Type**

Output:

Outlet_Size	Outlet_Type	Item_MRP
High	Supermarket Type1	141.425982
Medium	Supermarket Type1	140.297699
	Supermarket Type2	141.678634
	Supermarket Type3	139.801791
Small	Grocery Store	139.787088
	Supermarket Type1	142.589934

4. Frequency Table

Code:

```
[42]
✓ 0s
pd.crosstab(data['Outlet_Size'],data['Item_Type'], margins= False)
```

- **crosstab()** creates a **frequency table** (contingency table) that counts how many times each combination of **Outlet_Size** and **Item_Type** occurs in your data.

Output:

Item_Type	Baking Goods	Breads	Breakfast	Canned	Dairy	Frozen Foods	Fruits and Vegetables	Hard Drinks	Health and Hygiene	Household	Meat	Others	Seafood	Snack Foods	Soft Drinks	Starchy Foods
Outlet_Size																
High	73	25	13	65	80	92	142	23	61	103	41	16	5	125	49	19
Medium	203	83	36	217	218	274	413	75	170	289	149	52	21	408	137	48
Small	187	71	30	189	198	249	328	50	136	257	119	55	20	335	126	38